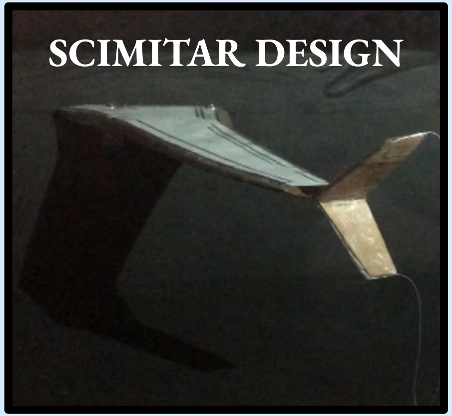
Which is the Most Efficient and Beneficial Winglet?



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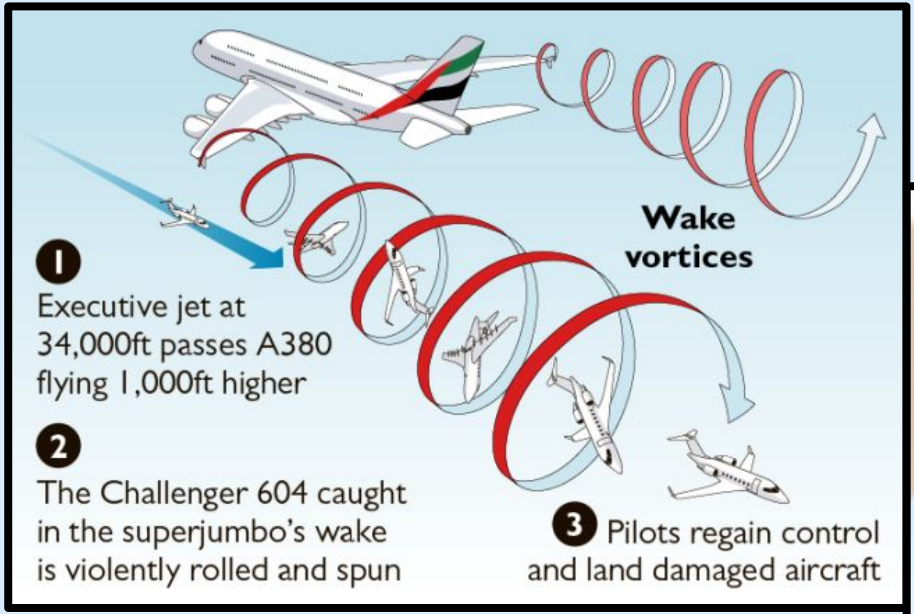
Quick Overview

Winglets are vertical extensions at wingtips. They are intended to improve the efficiency of fixed-wing aircraft by reducing drag. Although there are several types of wing tip devices, their intended effect is always to reduce an aircraft's drag by partial recovery of the tip vortex energy.

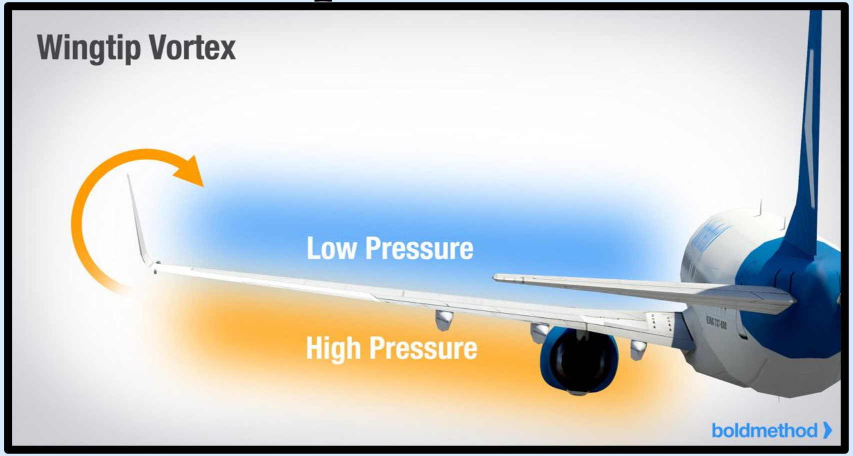
True Story

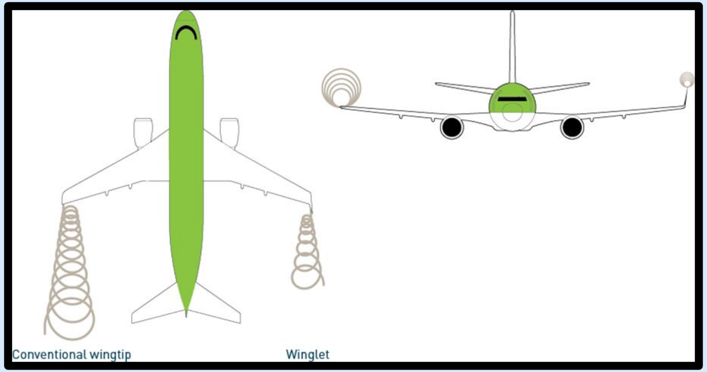


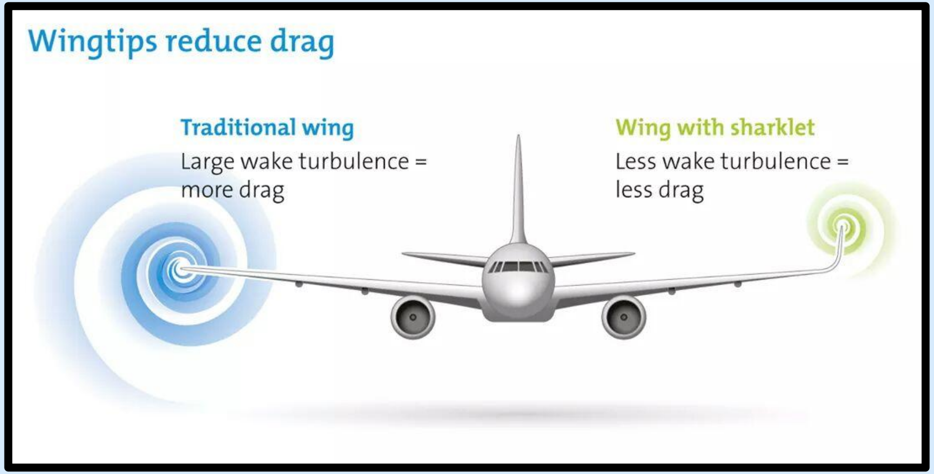
Wake vortices
1
Executive jet at 34,000ft passes A380 flying 1,000ft higher
2
The Challenger 604 caught in the superjumbo's wake is violently rolled and spun
3
Pilots regain control and land damaged aircraft

BEFORE
AFTER
FlightServiceBureau.org

Visual Representations Of Vortex (Wake Turbulence)







Problem(s)

There is a huge chain of problems which winglets help counter. Winglets have been proven to reduce aerodynamic drag as the airflow of a wing moves to the tip, higher pressured air from below the wing moves around and rises to the top. The circular flow forms a vortex at the end of the wing called dirty air. When this dirty air adds downwards pressure onto the wing, it reduces the lift it can generate by significant levels forcing the planes engines to work harder and burn more fuel to generate the needed lift. The more fuel burned means that more carbon dioxide, nitrogen oxide, and other harmful chemicals are being emitted into the atmosphere. Also, the more fuel burned faster means reduced range lessening the number of destinations a plane can reach. Winglets also help further stabilize flight and actually help reduce a planes sound by absorbing a lot of the sound waves produced by the loud process of the airfoil and engines as it occurs under high levels of stress and pressure. This allows for more comfortable flights and happier communities with less noise complaints down on the ground. Next, winglets help improve the problem of delayed flights as on average, a duration of 2 - 3 minutes is needed between each take-off to make sure that the plane behind another avoids wake turbulence which is the disturbance of the atmosphere formed by an aircraft as it passes through the air. Wake turbulence can actually be very dangerous and in some cases even powerful enough to completely disrupt a smaller planes flight pattern. Overall, there are many problems that winglets in general help solve but the question is, which winglet design helps solve these problems in a more effective manner.

Claim

I hypothesize that a split scimitar winglet would be the most efficient winglet design because not only does it have an upwards curve at the end of the wing, it also includes a symmetrical downwards curve which can further weaken and delay the high pressured airs rise to the top of the wing which significantly weakens the vortex energy.

Materials

1. **FOR WING**
   - Cardboard
   - Toothpicks
   - Gorilla Scotch Tape
   - String
   - Scissors
2. **REQUIRED FOR BUILDING TUNNEL**
   - 3 x (2 by 1 foot thin wood planks)
   - 1 x (2 by 1 foot plexiglass)
   - 20 x (small machine screws)
   - Screw Driver
   - Gorilla Scotch Tape
   - Plastic Wrap
   - Fan
   - Aluminum Foil
   - Utility Knife
3. **NOT REQUIRED**
   - Dry Ice and Hot Water (we only used this to see which way the results would be more clear and string on the wing showed better)
   - Black (Spray Paint)

First Design

Our first design or component was a normal airplane wing without any winglet or attachments to it.

This is the test wing without a winglet.

This is what a plane without any winglet would look like in real life and our model represents the function of this.

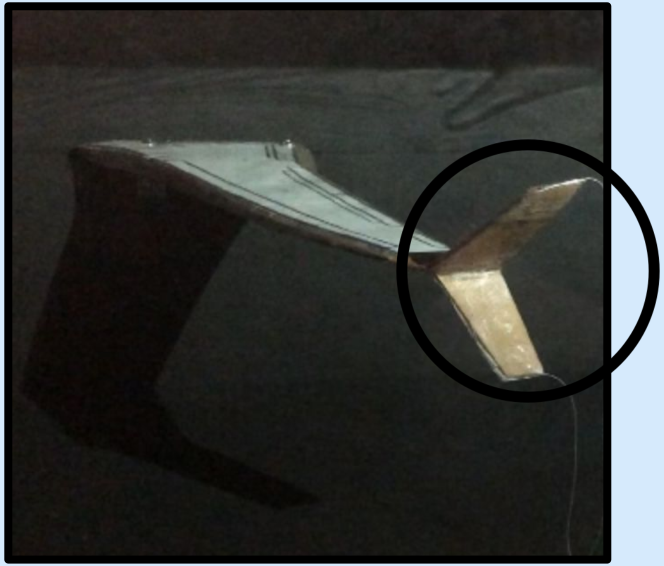
Second Design

This is the test wing with an attached sharklet winglet.

The main change that was made with the second design is that we have added a vertical winglet that goes up. The sharklet, getting its name from its similarities to a shark fin is placed to counter wake turbulence and the vortex created.

This is the sharklet winglet in real life which the model was built to represent.

Third Design



This is the test wing with an attached scimitar winglet.

The changes made in this third design were that an additional upside down sharklet symmetrical downwards to the top one was added. This design is called the scimitar winglet and the lower sharklets job is to further delay the high pressures rise and to further weaken the vortex created.

This is the real scimitar winglet which the test wing is supposed to represent.

Evidence/Data

The way the efficiency of the winglets was tested is that for each one, we attached a string to its top corner end and when put into the wind tunnel, we recorded the intensity of the height of the strings movements and the vibrations in the string as it was set right in the area where the vortex is created so by the size of the vibration in the string, we can tell whether or not it is a big or small vortex. Big obviously being that it is not that efficient while a small vortex means that the winglet was able to reduce the amount of dirty air being created once bouncing off the wing.

	Trial 1	Trial 2	Trial 3	Average height string moves up and down.
No Winglet	1.25 inches from its top most point to its bottom most point.	1.75 inches from its top most point to its bottom most point.	1.6 inches from its top most point to its bottom most point.	1.5333... inches from its top most point to its bottom most point.
Sharklet	1.55 inches from its top most point to its bottom most point.	1.3 inches from its top most point to its bottom most point.	1.5 inches from its top most point to its bottom most point.	1.45 inches from its top most point to its bottom most point.
Scimitar	0.8 inches from its top most point to its bottom most point.	1.2 inches from its top most point to its bottom most point.	0.95 inches from its top most point to its bottom most point.	0.9833... inches from its top most point to its bottom most point.

Evidence/Data: Each Wing's Wind Tunnel Trials

NO WINGLET STRING:
https://kapwi.ng/c/6fCmyETN

SHARKLET STRING:
https://kapwi.ng/c/nQMH23Ho

SCIMITAR WINGLET STRING:
https://kapwi.ng/c/HTnO3cFe

NO WINGLET VAPOR:
https://kapwi.ng/c/23IIjpLA

SHARKLET VAPOR:
https://kapwi.ng/c/k9kQ2shp

SCIMITAR WINGLET VAPOR:
https://kapwi.ng/c/USetEp43

Reasoning

My claim was that “I hypothesize that a split scimitar winglet would be the most efficient winglet design because not only does it have an upwards curve at the end of the wing, it also includes a symmetrical downwards curve which can further weaken and delay the high pressured airs rise to the top of the wing which significantly weakens the vortex energy.” As stated in the claim, it was predicted that the scimitar design would be the most efficient which turned out to be true. This claim turned out to be proven true by the series of tests conducted on each of the winglets. Stated in the table, the average height that the string had hit a top most point and a bottom most point for the wing without a winglet was 1.5333… inches. The average height for the wing with a sharklet was 1.45 inches and lastly, the average height of the wing with a scimitar winglet was 0.9833… inches. This evidence here proves just how much the scimitar winglet reduced the strings vibrations up and down. From no winglet which is approximately 1.53 inches to scimitar which is 0.9833 inches, we see a 35% reduction in the height of the vibrations. This here proves that indeed the scimitar winglet did reduce the size of the vortex as we got way weaker vibrations after the string was placed in the scimitar winglets vortex. This all contributes to the final conclusion that the scimitar winglet is the most efficient.

Reflection

Why does this project matter?

This project matters a substantial amount because it helps clear up many confusions and helps solve many problems. Now knowing which winglet design is the most efficient, we can base our decisions on our knowledge to further counter problems more effectively. The information this project has taught us can be used to counter many problems like reducing delayed flights, reducing harmful emissions, reducing sound and noise, and increasing lift which all benefit the environment and increase efficiency overall. In conclusion, this project helps contribute to the awareness of solving many problems at once by just altering a few design characteristics on a planes wingtip.

New questions/problems to solve?

1. How can we completely get rid of the vortex instead of just weakening it or shrinking it?
2. Is there a way to create a completely zero-emission engine?
3. Would smaller winglets on the horizontal stabilizers of a plane help?
4. Is there a way to make a noiseless engine and if not, can we replace engines with motors?
5. One problem that we had to deal with was the fact that since the fans suction was too strong, the dry ice's vapor would pass too quickly not revealing the wake turbulence that much which made us use the strings to counter this problem.

Conclusion

According to the results gathered by the extensive testing and trials on all three designs, we can conclude that the scimitar winglet design which is the third design is the most efficient and beneficial winglet out of the other choices and that airlines should look forward to purchasing aircraft with this option for maximized efficiency.

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- http://www.nycaviation.com/2014/05/turbulence-everything-need-know-part-ii#.VDpMkPmSySo
- https://www.boldmethod.com/learn-to-fly/aerodynamics/how-winglets-work-to-reduce-drag-and-how-wingtip-vortices-form/

Videos/ Documentaries:

- https://www.youtube.com/watch?v=FNqXf6t7e-w
- https://www.youtube.com/watch?v=vD828p9Nt0U
- https://www.youtube.com/watch?v=CnvIf3vFEYA
- https://www.youtube.com/watch?v=Lc86Akaq3KQ